

ZUL IKRAM MUSADDIK RAYAT

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Summary

Aspiring Software Engineer and final-semester Mechanical Engineering student at BUET with over 4 years of professional experience in full-stack development. Expert in building scalable, production-ready applications, having successfully architected systems supporting up to 80,000 users. Proven track record in AI integration, specifically in developing RAG pipelines and custom OCR models for Bangla text with 98% accuracy. Combines a rigorous mechanical engineering foundation with deep technical expertise in Nest.js, Python, and cloud infrastructure to deliver high-impact, performant software solutions.

Education

B.Sc in Mechanical Engineering (Final Semester) / *Bangladesh University of Engineering and Technology*

- CGPA: 3.08 (out of 4)
- Former Class Representative

Higher Secondary Certificate (HSC) / *Notre Dame College, Dhaka*

2020

- GPA: 5.00

Work Experience

Software Engineer - L1 (Part-time) / *RubizCode IT*

May 2025 – Sept 2025

- Leading development of scalable Ed-Tech and **AI-integrated** software.
- **Architected** cloud-based **microservices** using NestJS, Prisma, PostgreSQL, and Redis.
- **Incorporated real-time** features (comments, polls, live exams) with WebSockets and message queues.
- **Contributed** to the **DevOps** pipeline with Docker, Traefik, and Cloudflare for deployment and scaling.

Software Engineer - Web (Part-time) / *Biology Haters*

Dec 2023 - Jul 2024

- **Built** Taalaash, a Bangla academic search engine with **OCR + ML** to search across thousands of pages. Taalaash quickly became the most popular product of the company and remains so to this day.
- **Achieved** efficient **full-text search** and ranking using PostgreSQL and Algolia.
- **Enhanced** UI/UX with **Next.js**, React, and Tailwind for students and educators.
- Image to Bangla text extraction and text highlighting on image using **FastAPI** and Tesseract.
- Tuned Tesseract to be able to extract Bangla text from images with 98% accuracy.
- Made the system efficient up to 95% by adding CDN and data layer caching, and made the system scalable.

Junior Software Engineer (Part-time) / *Ronon (ACS EdTech)*

Dec 2021 - Dec 2023

- **Designed** the core engine of Ronon, a **real-time** doubt-solving platform. It quickly grew with a stunning 20 thousand users and 5-10 thousand monthly active users. Currently, Ronon has a userbase of 80 thousand users.
- Founded and led the development of Matrahin, an interactive physics simulation tool blending design, development, and engineering.
- Collaborated with educators to build scalable edtech solutions in Python (Django), Node.js, and React.

Extracurricular Activities

President / *BUET Animata - Animation & VFX Club*

Mar 2025 – Present

- Served as Founding Assistant General Secretary; organized workshops on basic animation and 3D modeling.
- Designed and animated the official club logo and visual assets.

Web Master / *Team Interplanetar - BUET Mars Rover Team*

Oct 2022 – Oct 2024

- Developed and maintained the home website of Team Interplanetar containing detailed descriptions of its products, findings and achievements.
- Forged a new certifications system in the website to provide certifications on engineering aspects of mars rovers.

Skills

Languages: JavaScript, TypeScript, Python, Go, C/C++, C#, HTML, CSS

Frontend: React, Next.js, Redux, Svelte, Tailwind CSS, Framer Motion, React-Query, Jotai, Zustand, Vite

Backend: Node.js, Nest.js, Express.js, Hono, FastAPI, Django, .NET Core

AI & Machine Learning: RAG (Retrieval-Augmented Generation), LangChain, Pinecone, OpenAI API, Tesseract OCR, TensorFlow, Scikit-Learn, NumPy, Pandas

Databases & Caching: PostgreSQL, MySQL, MongoDB, Redis (Streams, Pub/Sub), Cloud Firestore, Pinecone (Vector DB).

DevOps & Infrastructure: Docker, Docker Swarm, Traefik, Cloudflare, Github Actions, AWS, Google Cloud Platform (GCP), Vercel, Heroku.

Tools & Middleware: RabbitMQ, Celery, FFmpeg, Git, Linux, Webhooks, gRPC, LaTeX.

Projects

TopperOn Doubts AI

RubizCode

- Created an AI-powered student doubt solver.
- Pioneered a full RAG pipeline utilizing a vector database and embeddings, decreasing average query latency by 35% and improving context relevance scores by an average of 0.2 points.
- Technologies: Python, FastAPI, **LangChain**, Pinecone, OpenAI API, RAG, **Next.js**, NestJS, Redis, **PostgreSQL**, GCP, CI/CD

Taalaash - Book Search Engine

Biology Haters

- Indexed 20,000+ textbook pages with OCR + ML, reaching 30,000+ students and 2,000+ daily searches.
- Achieved 98% OCR accuracy for Bangla with custom-tuned Tesseract models.
- Included with AI-based reranking (Pinecone), improving search relevance by 35% and reducing indexing time by 80%.
- Scaled backend to support 100,000+ users and millions of queries with 99.9% uptime.
- Technologies: Python, FastAPI, Tesseract, Node.js, Next.js, Responsive Design, **PostgreSQL**, **GCP**, **Pinecone**, CI/CD

Ronon Doubts

Ronon

- Currently handling 80,000+ registered users with 5,000–10,000 monthly active users within 2 years.
- Integrated LaTeX rendering, improving clarity for math/physics answers by 70%.
- Optimized backend with Django + PostgreSQL, handling 15,000+ concurrent queries daily.
- Boosted question resolution rate to 95% within 24 hours.
- Technologies: Python, Django, **PostgreSQL**, React, LaTeX, **Tailwind**

Video Transcoding Pipeline

Personal

- Real-world **system design** (multi-service, infra, failure modes)
- **Message queues & streaming** with Redis Streams and consumer groups
- Advanced **Redis** usage (streams, pub/sub, caching)
- **FFmpeg** and practical video processing pipelines
- End-to-end integration between **Go backend** and a **React/Vite** dashboard

Achievements & Certifications

- Introduction to Front-End Development | *Coursera*
- Networking in Google Cloud: Fundamentals | *Google*
- SQL for Data Science | *Coursera*
- SQL (Advanced) | *HackerRank*
- Learn RAG | *Scrimba*

Reference

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Professor

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Md. Tareq Hasan Siddiqui

MD

Ronon

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